

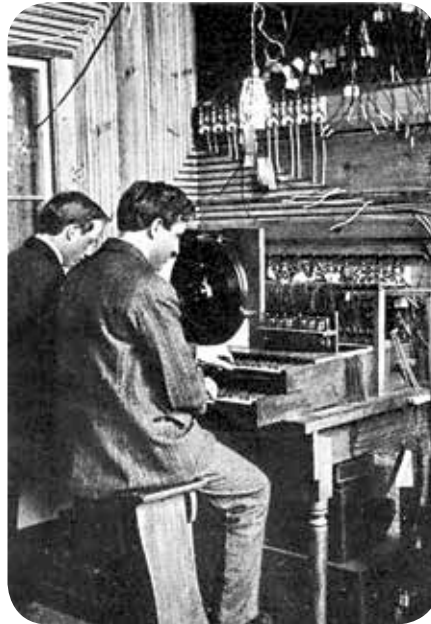
The Telharmonium

If you thought the Moog Synthesizer or Hammond Organ were the first musical instruments that could synthesize music, you couldn't be more wrong. That honour is often given to the Telharmonium, a device created by Thaddeus Cahill way back in 1897. It used tone-wheels to generate musical sounds as electrical signals by additive synthesis. It created music through electrical signals which were transmitted by wire to a loudspeaker for playback. In fact Cahill was convinced he could transmit music over telephone to remote locations. Early public exhibitions of the grand instrument gathered people in hordes. And yet there was almost a 50 year gap between the Telharmonium and the first mass produced electronic musical instrument. Why didn't this technology carry on? For one thing the Telharmonium was way ahead of its time. To impede its way to mainstream adoption were further hurdles: the Telharmonium was huge; literally one of the biggest instruments ever built. It weighed 200 tons, was the size of a room and here's what rang it's death knell quicker than anything else – it cost a whopping \$200,000 to build way back then!

Cahill did manage to transmit music over the telephone, however the signal would leak into other calls and the experiment proved futile. Only three of these machines were ever made, and today not a sprocket remains of the original three.

Valyrian Steel err Damascus Steel

Despite the Game of Thrones reference, real world legends really do speak of blades from Persia forged from a type of metal that could cut swords cleanly in half. These razor sharp swords were made from Damascus Steel, and were known for their strength and cutting prowess. These blades had a peculiar swirling pattern (like flowing water) on their surface and were feared on the battlefield. Stories of particular swords were shared over campfires by weary soldiers and their legend grew. What was Damascus



Behold the 200 ton sound producing behemoth

Steel? It was an alloy created using wootz steel imported from India. Modern-day analysis of these swords has found some startling facts. A study in Germany found carbon nanotubes in a blade forged from Damascus steel! So we're talking of sharpness that measures an atom thin way back in the 1700s. And as meteoric as this rise to eminence was, the exact process of forging Damascus Steel is lost. Experts believe the sword's strength came from a process which moulded cementite and soft iron to create a metal that was both strong and flexible. Theories suggest that the technique was not an exact science to begin with and that only a small part of the swords made by sword makers fit into the "Damascus Steel" standard. Other theories suggest that raw materials were running low and the original sword-smiths just abandoned the process. Whatever the reason, modern-day attempts at making Damascus Steel have proved futile and this technology will forever remain lost.

NASA's space race technology

The Gemini and Apollo space programs are what eventually put man on the moon. The Gemini program provided

the groundwork needed for manned space flight (1965-66) and the Apollo missions took it ahead beyond the stratosphere. If you'll notice, these are events that took place relatively recently so the times that we refer to here are not strictly lodged in antiquity. Despite that, the space race was so frantic that much of the technology developed during that time didn't have any records; they were in that much of a hurry to reach the moon. In many cases private contractors were called in to assist with different modules and not every detail/process was recorded for posterity. The technology is not strictly lost – NASA does have a bunch of Saturn V rockets on their campus (or in museums) but very few of the exact schematics and plans for the original parts are available today. If NASA wants to renew the space program and launch a mission to the moon,



Remnants of the space race

a lot of the tech will have to be reverse engineered, and reportedly NASA is doing exactly that. What we described here is of course a near impossibility now thanks to digital workflows, infinite storage capacity and the internet. But it's startling how something so recent can be lost, which makes the older accounts of lost tech all the more believable...